Kinase Library motif enrichment

Which kinases the measured changes are consistent with — prepared here, computed elsewhere, and drawn here again.

AT A GLANCE

Where	Phospho → Kinase Library, headed Kinase Library Motif Enrichment. The tab is not shown in viewer mode.
Data level	Phosphosite.
Needs	An annotated phospho dataset with UPD_seq, two different conditions (four under Volcano Control), and — for the figure — two result files from the external tool.
Substrate sets	None are stored here: the motif and substrate definitions belong to the external kinase library the page sends you to.
Score	That tool's frequency factor, plotted as its base-2 logarithm.
Significance	The adjusted p-value in the uploaded file; no correction is applied here.
New	Not covered by any previous manual.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the enrichment infers, and what this page does

This page prepares input for the Kinase Library's motif-based Fisher enrichment workflow and visualises its returned results. The plot is not a direct measurement of kinase activity. Motif enrichment generates hypotheses about signalling, which also depends on substrate availability, phosphatases, protein abundance and experimental coverage. This is not kinase-

substrate enrichment analysis (KSEA, Casado et al. 2013) or another substrate-based z-score approach; their scores and input requirements are not interchangeable with this frequency-factor plot. CoPYlot does not run the external enrichment. The workflow has three steps:

1. **Prepare the input.** A phosphosite differential-abundance table is computed from your data and formatted for the external tool, each modified sequence re-centred on its modified residue, and downloads as TSV.
2. **Run the enrichment elsewhere.** The page links to the external kinase library's Fisher enrichment analysis, where the table is uploaded twice — once for serine/threonine kinases, once for tyrosine and non-canonical — with `UPD_seq` as the motif column, `log2FC` as the effect column and `pval` or `adj_pval` as the optional p-value column.
3. **Draw the results.** The two result files are uploaded back here and drawn as one volcano of kinases. Matching column names alone do not make results from another method comparable: verify its score definition, contrast and correction before importing it.

> **READ THE SCORE, NOT A KINASE FOLD CHANGE**
>
> A frequency factor of 4 has $\log_2$ value 2. In the down-regulated-site result it is placed on the negative side, so the side identifies the submitted site set. This is enrichment among that set, not a fourfold change in kinase abundance or a direct measurement of its activity. An adjusted p-value of 0.01 has height 2 on the vertical axis.

Reading it

- One point, one kinase. The two files are concatenated, so a kinase reported by both appears twice.
- The x axis, $\log_2$(frequency factor). Positive means the kinase's substrate set is enriched among the sites that rose in **Condition 1**; negative, among those that fell.
- The y axis, $-\log_{10}$ of the adjusted p-value the result file gives. A p-value of zero is drawn at $-\log_{10}(10^{-300})$ rather than at infinity.
- The dashed green line at $-\log_{10}$ of **Kinase Library Volcano P-value Threshold**. Points above it are red or blue by the sign of x; the rest are light grey, `Not significant`.
- How many substrates a point rests on is *not* on the figure. A substrate-count column is read and validated but neither plotted nor shown in the hover.

> **EASY TO GET WRONG**
>
> The marker size does not express the strength of biological evidence. Inspect the contributing motifs, counts where supplied, effect size, multiple-testing procedure and experimental context together. Neither a large substrate set nor a small p-value alone establishes kinase activity.

2 Controls

Controls on this page: the input table

Control	What it does	Default	Where
Source	Volcano for a plain two-condition comparison, Volcano Control to subtract a control from each side first; the latter requires four different conditions.	Volcano	page
Data Source	Which available matrix the comparison runs on: Data, Filtered Data or Imputed Data. Check the selection before exporting.	current selection	page
Condition 1, Condition 2	The two conditions compared; $\log_2$ FC = **Condition 1** − **Condition 2**. The same one twice produces no table.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only; each is subtracted from its own side first.	first condition in the annotation	page
P-value Threshold, log2 FC Threshold	Together they set the significance label in the exported table: below the one in adjusted p and above the other in magnitude. Neither removes a row. The p-value is clamped to $[10^{-12}, 1]$.	0.05, 1	page
Test Type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples). Volcano Control offers instead Unpaired (propagated variance) and Paired (propagated control variance), promoting an equal-variance choice to the Welch form: the control means add uncertainty of their own.	Unpaired (Student's t)	page
Motif Window Proteome	Human, Mouse, or no proteome. Protein sequence supplies flanking residues where the site can be matched; fallback peptide windows are reported.	Human	page
Use Uncorrected P-values	Selects raw pval for selected_pval and classification. adj_pval remains Benjamini–Hochberg adjusted.	off	page
Download Kinase Library Table (TSV)	Saves the table, named for the comparison.	—	page

Controls on this page: the volcano from uploaded results

Control	What it does	Default	Where
Serine/Threonine Result File, Tyrosine/Non-canonical Result File	The two files that come back from the external tool. .csv, .tsv or .txt. Both are required.	none	page
Kinase Library Volcano P-value Threshold	The adjusted p-value below which a kinase is coloured, and where the green line is drawn. Separate from **P-value Threshold** above.	0.1	page
Annotate Hits	Prints the kinase name beside every point that is not Not significant.	off	page
Generate Kinase Library Volcano	Builds the figure, once both files are present and the conditions validly chosen.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Reference line	The significance line: shown or hidden, and its colour, dash pattern and width.	shown, green, dashed, 1.5	appearance
Title, axis titles, legend	Text, size, colour, and whether each is drawn. The plot writes Kinase Library Motif Enrichment: cond1 vs. cond2 and names the contrast in the x title.	shown	appearance
Points	Marker size, opacity, outline and shape of the interactive volcano. The colour stays the regulation class's, so the card has no colour field here.	size 10, opaque, black outline 0.5	appearance
Background, theme	Paper colour, preset themes.	default	appearance

An imputed phosphosite matrix is allowed. It changes the evidence entering the comparison: inspect the reported imputation counts and retain the settings rather than treating filled values as additional observations.

> **WORTH KNOWING**
>
> The third table is the plot appearance panel, hidden until advanced mode is on (**Ctrl** + **Shift** + **E**) and described in Part XI.

3 How it is computed

THE INPUT TABLE

Every phosphosite is tested between the two conditions exactly as the volcano plot tests it, and with the direction the volcano plot uses:

$$\log_2 \mathrm{FC} \;=\; \overline{x}_{\text{cond 1}} - \overline{x}_{\text{cond 2}},$$

or, under Volcano Control, $(\overline{x}_{\text{cond 1}} - \overline{x}_{\text{ctrl 1}}) - (\overline{x}_{\text{cond 2}} - \overline{x}_{\text{ctrl 2}})$: the same difference, taken after each side has had the mean of its own control subtracted from it.
UPD_seq contains a fifteen-residue window: the modified residue and seven positions on each side. With a selected proteome, the site position and peptide residues are checked against the protein sequence. If this succeeds, only positions beyond the protein terminus are padded with underscores. Otherwise a peptide-derived window is used and window_source says peptide. Inspect these fallbacks: a peptide end is not necessarily a protein end, and missing flanks can affect motif scoring. Every row of the comparison is exported; the thresholds only decide what each row's significance column says.

THE VOLCANO FROM THE UPLOADED RESULTS

Both result files are read and stacked. Two quantities are required per row, found by column name, ignoring case, punctuation and a subscript 2:

- the effect, from Log2(Frequency Factor) taken as it stands, or from Frequency Factor, which must be positive and whose base-2 logarithm is then taken;
- the significance, from Adjusted p-value, which must lie in $[0, 1]$.

A kinase name is taken from Kinase, Gene Name or Gene, a substrate count from Substrate Count, Number of Substrates or n substrates when one is present. Two columns that mean the same thing and disagree are an error rather than a silent choice, and so is a row missing either required quantity.
The figure is then arithmetic and nothing else: $y = -\log_{10}(p)$ with p clipped up to 10^{-300}; a point is Enriched in up-regulated sites if $p <$ **Kinase Library Volcano P-value Threshold** and the effect is positive, Enriched in down-regulated sites if $p <$ threshold and the effect is negative, Not significant otherwise; and a line is drawn at $-\log_{10}$ of the threshold. No p-value is adjusted here, no score normalised, no substrate set consulted.

Under Volcano Control the contrast computed, and printed on the x axis, is (cond 1 − ctrl 1) − (cond 2 − ctrl 2). Keep that orientation when interpreting the external results.

Edge cases

- A paired test needs every sample paired before either the table or the figure appears, and Volcano Control rejects any selection in which the four conditions are not distinct.
- Uploaded files with no rows, duplicate column names, a non-positive frequency factor or an adjusted p-value outside $[0, 1]$ are refused with a message naming the problem.
- A kinase in both result files is drawn twice; the two runs cover disjoint families, so an overlap is worth investigating.

Nothing records which release of the external library a set of result files came from, and the session does not keep the files themselves. Keep the downloaded result files with the rest of the analysis: they are the only record of what the enrichment was run against.

4 The input table

Columns in this table

Column	What it holds
UPD_seq	The re-centred motif — the column the external tool reads as the motif.
log2FC	The fold change of that site, in the direction stated above.
pval, adj_pval	The raw p-value and its Benjamini–Hochberg adjusted form, stored separately.
selected_pval	The raw or adjusted value chosen for classification.
significance	Upregulated, Downregulated, Not significant or Unavailable, from the two thresholds. Carried for reference only.
n_pairs, neg_log10_selected_pval	The usable pairs behind that site, on paired tests only, and $-\log_{10}$ of the selected p-value.
Protein, Gene, Site, PTM_Collapse_key, window_source	Site identity and whether the motif window came from the protein sequence or from the peptide fallback.

The table downloads as tab-separated text. The figure has no image download: it is interactive, and saves from its own toolbar.

There is no required number of significant kinases. No hits can reflect limited evidence, coverage, thresholds or genuine lack of enrichment; many hits can reflect related motifs as well as broad changes. The plot alone does not distinguish these explanations.

Upload the full eligible site comparison, not only its significant rows: the external method needs the comparison population to define its groups. Choose an optional p-value column if you intend the external classification to require both effect and statistical support. Omitting it

makes that classification depend on effect alone; it does not remove the external enrichment test. State whether the supplied values were raw or adjusted and record the external thresholds. The notebook download beside this table reproduces preparation of the export only, not the external analysis or its returned kinase plot.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

The two steps of the page are two functions, with the trip to the external tool in between. kinase_library_data_f builds the input table, and kinase_library_volcano draws the result files once they come back:

PYTHON

```
kl_input = kinase_library_data_f(data3, meta2, "WT", "GRK")
fig = kinase_library_volcano(read_data("kl_st_results.csv"),
                             read_data("kl_tnc_results.csv"),
                             condition1="WT", condition2="GRK")
```

R

```
kl_input <- kinase_library_data_f(data3, meta2, "WT", "GRK")
fig <- kinase_library_volcano(read_data("kl_st_results.csv"),
                              read_data("kl_tnc_results.csv"),
                              condition1 = "WT", condition2 = "GRK")
```

The arguments of the first are the controls of the input-table card: in_pval and in_log2fc are the two thresholds, uncorrected is **Use Uncorrected P-values**, and paired and test_type carry **Test Type**; kinase_library_data_f_control is the Volcano Control source, taking the two control conditions as well. In kinase_library_volcano, pval_thresh is **Kinase Library Volcano P-value Threshold** and defaults to 0.1 as on the page. Neither function performs the enrichment: the two frames handed to kinase_library_volcano are the files the external tool produced, and the notebook does no more than the page does with them. The older names ksea_data_f, ksea_data_f_control and ksea_volcano still exist in both languages as aliases of these three, so existing notebooks keep working.

6 Further reading

The Kinase Library's official repository documents its motif-based kinase prediction and enrichment workflow. This is the relevant external implementation for the imported frequency-factor results, not an interchangeable label for every method called KSEA.

Behind this page

Function	What it produces
ksea_table	The prepared input table, standard or control-corrected. The application's function keeps the older name
kinase_library_data_f, kinase_library_data_f_control	The same table in the notebook packages, plain and control-corrected; ksea_data_f and ksea_data_f_control are aliases
ksea_uploaded_volcano_html	The interactive volcano built from the two result files. The application's function keeps the older name
kinase_library_volcano	The equivalent figure in the notebook packages; ksea_volcano is an alias